

Linear Algebra W244

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Note for the student

These lecture notes for Linear Algebra W244 follow on directly from the lecture notes for Linear Algebra W214, which ended with Chapter 4.3. Diagonalizing matrices.

Chapter 4

Eigenvalues and eigenvectors

4.3 Diagonalizing matrices

Definition 4.3.1. We say that an $n \times n$ matrix A is *diagonalizable* if there exists an invertible $n \times n$ matrix P such that $P^{-1}AP$ is a diagonal matrix.

Example 4.3.1. $A = \begin{bmatrix} 2 & 1 \\ \frac{1}{2} & \frac{5}{2} \end{bmatrix}$ is diagonalizable, since if we set

$$P = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}$$

then

$$P^{-1} = -\frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & -2 \end{bmatrix}$$

and we check,

$$\begin{aligned} P^{-1}AP &= -\frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ \frac{1}{2} & \frac{5}{2} \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \\ &= -\frac{1}{3} \begin{bmatrix} \frac{3}{2} & -\frac{3}{2} \\ -3 & -6 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \\ &= -\frac{1}{3} \begin{bmatrix} -\frac{9}{2} & 0 \\ 0 & -9 \end{bmatrix} \\ &= \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 3 \end{bmatrix} \end{aligned}$$

which is indeed a diagonal matrix.

Lemma 4.3.2 (Characterization of Diagonalizable Matrices). *Let A be an $n \times n$ matrix. The following are equivalent:*

1. A is diagonalizable.
2. There exists a basis of column vectors $\{w_1, \dots, w_n\}$ of Col_n consisting of eigenvectors of A .

Proof. (1) $\Rightarrow$ (2). Suppose $P^{-1}AP = D$, where

$$D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

is a diagonal matrix. Let $w_i = Pe_i$, where

$$e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

is the i th basis vector in the standard basis for Col_n . Then $\{w_1, \dots, w_n\}$ is a basis for Col_n (Exercise 4.3.1). Moreover,

$$\begin{aligned} Aw_i &= APe_i \\ &= PDe_i && (\text{since } P^{-1}AP = D \text{ means } AP = PD) \\ &= P\lambda_i e_i \\ &= \lambda_i Pe_i && (\text{by Proposition 69 from W214}) \\ &= \lambda_i w_i \end{aligned}$$

so w_i is an eigenvector of A with eigenvalue λ_i .

(2) $\Rightarrow$ (1). Suppose there exists a basis of column vectors $\mathcal{C} = \{w_1, \dots, w_n\}$ of Col_n consisting of eigenvectors of A . Define a matrix P by assembling the column vectors w_i into a matrix:

$$P := \begin{bmatrix} \boxed{w_1} & \boxed{w_2} & \dots & \boxed{w_n} \end{bmatrix}.$$

Then P is invertible, since we can interpret P as the change-of-basis matrix from the basis $\mathcal{C} = \{w_1, \dots, w_n\}$ to the standard basis $\mathcal{B} = \{e_1, \dots, e_n\}$:

$$P = [\text{id}]_{\mathcal{B} \leftarrow \mathcal{C}}$$



Do you understand this? If w is a column vector, and $\mathcal{B} = \{e_1, \dots, e_n\}$ is the standard basis for Col_n , then $[w]_{\mathcal{B}} = w$! Convince yourself of this.

and hence (by Corollary 45 of W214), P is invertible and in fact $P^{-1} = [\text{id}]_{\mathcal{C} \leftarrow \mathcal{B}}$. That is,

$$\begin{aligned} Pe_i &= w_i & i &= 1 \dots n \\ P^{-1}w_i &= e_i & i &= 1 \dots n. \end{aligned}$$

Moreover,

$$\begin{aligned} &\text{the } i\text{th column of } P^{-1}AP \\ &= P^{-1}APe_i && \text{(by Exercise 82 from W214)} \\ &= P^{-1}Aw_i \\ &= P^{-1}\lambda_i w_i \\ &= \lambda_i P^{-1}w_i && \text{(by Proposition 69 from W214)} \\ &= \lambda_i e_i \\ &= \text{the } i\text{th column of } D. \end{aligned}$$

Therefore, $P^{-1}AP = D$. □

Now answer the following questions experimentally, using the Desmos worksheet on eigenvectors and eigenvalues:

<https://www.desmos.com/calculator/dg2ubws4cn>

Example 4.3.2. Is

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

diagonalizable?

Solution. No, by experimenting in the Desmos worksheet, we see that although

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

is an eigenvector, there are not two linearly independent eigenvectors. So A cannot be diagonalizable, by Lemma 4.3.2